

Robust Portfolios

Portfolio Optimization — Chapter 14

Daniel P. Palomar (2025). *Portfolio Optimization: Theory and Application*.
Cambridge University Press.

portfoliooptimizationbook.com

Outline

- 1 Introduction
- 2 Robust Portfolio Optimization
 - Robust Optimization
 - Robust Worst-Case Portfolios
 - Numerical Experiments
- 3 Portfolio Resampling
 - Resampling Methods
 - Portfolio Resampling
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- 4 Summary

Executive Summary

- Mean-variance portfolio optimization requires an estimation of the mean vector and covariance matrix.
- Estimation errors lead to unstable portfolio solutions that are highly sensitive to input parameters.
- This issue has been referred to as “Markowitz optimization enigma” and makes the approach less popular among practitioners.
- Two main solutions are overviewed in these slides (Palomar 2025, chap. 14):
 - **robust optimization**: takes into account parameter uncertainty
 - **resampling/bootstrapping**: aggregates solutions from resampled data

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Markowitz's Mean-Variance Portfolio: Trade-off between expected return $\mathbf{w}^T \boldsymbol{\mu}$ and risk measured by variance $\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \mathbf{w}^T \boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \\ & \text{subject to} && \mathbf{w} \in \mathcal{W} \end{aligned}$$

where

- parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ represent expected returns and covariance matrix
- λ is a hyper-parameter controlling investor's risk aversion
- $\mathcal{W}$ denotes constraint set, e.g., $\mathcal{W} = \{\mathbf{w} \mid \mathbf{1}^T \mathbf{w} = 1, \mathbf{w} \geq \mathbf{0}\}$.

Parameter Estimation:

- Parameters $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ are unknown and estimated using historical data $\mathbf{x}_1, \dots, \mathbf{x}_T$.
- Estimators range from simple sample estimators to sophisticated shrinkage heavy-tailed maximum likelihood estimators.
- Estimation error depends on the number of observations.
- Limited historical data and lack of stationarity lead to noisy estimates $\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$.
- Noisy estimates particularly affect $\hat{\boldsymbol{\mu}}$ (Michaud 1989; Chopra and Ziemba 1993).

Achilles' Heel of Portfolio Optimization:

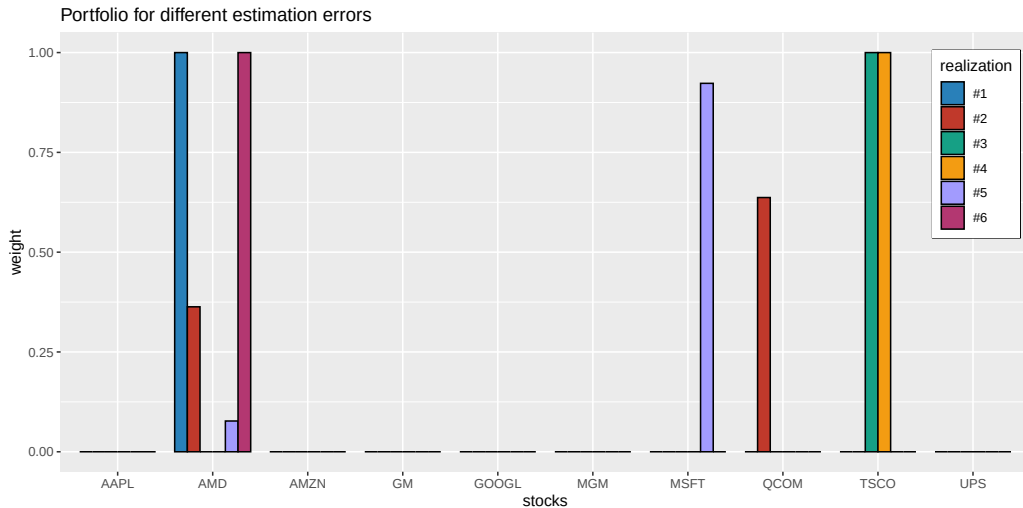
- Estimation noise in $\hat{\mu}$ and $\hat{\Sigma}$ leads to erratic portfolio designs.
- Known as “Markowitz optimization enigma” (Michaud 1989):
 - Portfolio optimization problems are “estimation-error maximizers.”
 - “Optimal” portfolios are financially meaningless (absence of significant investment value).

Sensitivity Illustration:

- Next figure shows sensitivity of mean-variance portfolio for six different realizations of estimation error.
- Behavior is erratic due to sensitivity to parameter errors.
- Each realization differs significantly, which is impractical for portfolio allocation.

Introduction

Sensitivity of the naive mean–variance portfolio:



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General Mathematical Optimization Problem:

(with optimization variable $\mathbf{x}$ and parameter θ)

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & f_0(\mathbf{x}; \theta) \\ \text{subject to} & f_i(\mathbf{x}; \theta) \leq 0, \quad i = 1, \dots, m \\ & h_i(\mathbf{x}; \theta) = 0, \quad i = 1, \dots, p, \end{array}$$

where

- f_0 : objective function
- f_i , $i = 1, \dots, m$: inequality constraint functions
- h_i , $i = 1, \dots, p$: equality constraint functions
- θ : external parameter (e.g., $\theta = (\mu, \Sigma)$ in mean-variance portfolio)
- solution denoted by $\mathbf{x}^*(\theta)$

Parameter Estimation:

- θ is unknown and estimated as $\hat{\theta}$.
- Solution using estimated parameter: $\mathbf{x}^*(\hat{\theta})$.
- Different from desired solution: $\mathbf{x}^*(\hat{\theta}) \neq \mathbf{x}^*(\theta)$.
- Question of approximate equality: $\mathbf{x}^*(\hat{\theta}) \approx \mathbf{x}^*(\theta)$.
- For mean-variance portfolio, solutions can be quite different as previously seen.

Making the Problem Robust to Parameter Errors:

• Stochastic optimization:

- Relies on probabilistic modeling of the parameter (Prekopa 1995; Ruszczyński and Shapiro 2003; Birge and Louveaux 2011).
- Includes: expectations (average behavior) and chance constraints (probabilistic constraints).

• Worst-case robust optimization:

- Relies on defining an uncertainty set for the parameter (Ben-Tal, El Ghaoui, and Nemirovski 2009; Ben-Tal and Nemirovski 2008; D. Bertsimas and Caramanis 2011; Lobo 2000).

Stochastic Robust Optimization:

- Estimated parameter $\hat{\theta}$ modeled as a random variable fluctuating around true value θ .
- True value modeled as:

$$\theta = \hat{\theta} + \delta,$$

where δ is the estimation error, a zero-mean random variable (e.g., Gaussian distribution).

- Importance of choosing the correct covariance matrix (or shape) for the error term.

Average Constraint:

- Instead of using the “naive constraint”:

$$f(\mathbf{x}; \hat{\theta}) \leq \alpha$$

use the “average constraint”:

$$\mathbb{E}_{\theta} [f(\mathbf{x}; \theta)] \leq \alpha.$$

- Interpretation: constraint satisfied on average, a relaxation of the true constraint.
- Preserves convexity: if $f(\cdot; \theta)$ is convex for each θ , so is its expected value over θ .

Stochastic Optimization: Implementation

- **Brute-Force Sampling:** Sample S times the random variable θ and use the constraint:

$$\frac{1}{S} \sum_{i=1}^S f(\mathbf{x}; \theta_i) \leq \alpha.$$

- **Adaptive Sampling:** Sample the random variable θ efficiently at each iteration while solving the problem.
- **Closed-Form Expression:** Compute the expected value in closed form when possible.
- **Stochastic Programming:** Various numerical methods developed for stochastic optimization (Prekopa 1995; Ruszczyński and Shapiro 2003; Birge and Louveaux 2011).

Example: Stochastic Average Constraint in Closed Form

- Quadratic constraint $f(\mathbf{x}; \boldsymbol{\theta}) = (\mathbf{c}^\top \mathbf{x})^2$, with parameter $\boldsymbol{\theta} = \mathbf{c}$.
- Modeled as $\mathbf{c} = \hat{\mathbf{c}} + \boldsymbol{\delta}$, where $\boldsymbol{\delta}$ is zero-mean with covariance matrix $\mathbf{Q}$.
- Expected value:

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\theta}} [f(\mathbf{x}; \boldsymbol{\theta})] &= \mathbb{E}_{\boldsymbol{\delta}} \left[\left((\hat{\mathbf{c}} + \boldsymbol{\delta})^\top \mathbf{x} \right)^2 \right] \\ &= \mathbb{E}_{\boldsymbol{\delta}} \left[(\hat{\mathbf{c}}^\top \mathbf{x})^2 + \mathbf{x}^\top \boldsymbol{\delta} \boldsymbol{\delta}^\top \mathbf{x} \right] \\ &= (\hat{\mathbf{c}}^\top \mathbf{x})^2 + \mathbf{x}^\top \mathbf{Q} \mathbf{x}\end{aligned}$$

- Interpretation: the robust expected value has the form of the naive version plus a quadratic regularization term.

Limitations of Average Constraints:

- No control over specific realizations of the estimation error.
- Constraint may be violated in many instances.
- Worst-case approach addresses this but can be too conservative.

Chance Constraints:

- Compromise between being too relaxed or overly conservative.
- Focus on satisfying the constraint with high probability (e.g., 95%).
- Replace naive constraint with:

$$\Pr[f(\mathbf{x}; \boldsymbol{\theta}) \leq \alpha] \geq \epsilon,$$

where ϵ is the confidence level (e.g., $\epsilon = 0.95$ for 95%).

- Generally hard to deal with, often requiring approximations (Ben-Tal and Nemirovski 2008; Ben-Tal, El Ghaoui, and Nemirovski 2009).

Worst-Case Robust Optimization

Worst-Case Robust Optimization:

- Parameter θ assumed to lie within an uncertainty region near the estimated value:

$$\theta \in \mathcal{U}_\theta,$$

where $\mathcal{U}_\theta$ is the uncertainty set centered at $\hat{\theta}$.

- Critical to choose the shape and size of the uncertainty set.

Typical Choices for Uncertainty Set Shape (Size ϵ):

- Spherical set:**

$$\mathcal{U}_\theta = \left\{ \theta \mid \|\theta - \hat{\theta}\|_2 \leq \epsilon \right\}$$

- Box set:**

$$\mathcal{U}_\theta = \left\{ \theta \mid \|\theta - \hat{\theta}\|_\infty \leq \epsilon \right\}$$

- Ellipsoidal set:**

$$\mathcal{U}_\theta = \left\{ \theta \mid (\theta - \hat{\theta})^\top \mathbf{S}^{-1} (\theta - \hat{\theta}) \leq \epsilon^2 \right\},$$

where $\mathbf{S} \succ \mathbf{0}$ defines the shape of the ellipsoid.

Worst-Case Robust Optimization

Worst-Case Constraint:

- Instead of using:

$$f(\mathbf{x}; \hat{\boldsymbol{\theta}}) \leq \alpha,$$

use the worst-case constraint:

$$\sup_{\boldsymbol{\theta} \in \mathcal{U}_{\hat{\boldsymbol{\theta}}}} f(\mathbf{x}; \boldsymbol{\theta}) \leq \alpha.$$

- Interpretation: constraint satisfied for any point inside the uncertainty set, a conservative approach.
- Preserves convexity: if $f(\cdot; \boldsymbol{\theta})$ is convex for each $\boldsymbol{\theta}$, so is its worst-case over $\boldsymbol{\theta}$.

Variation: Distributionally Robust Optimization:

- Applies worst-case uncertainty philosophy to probability distributions.
- Referred to as distributional uncertainty models or distributionally robust optimization (D. Bertsimas and Caramanis 2011).

Worst-Case Robust Optimization: Implementation

- **Brute-Force Sampling:**

- Sample S times the uncertainty set $\mathcal{U}_\theta$ and use the constraint:

$$\max_{i=1,\dots,S} f(\mathbf{x}; \theta_i) \leq \alpha$$

- Or equivalently, include S constraints:

$$f(\mathbf{x}; \theta_i) \leq \alpha, \quad i = 1, \dots, S;$$

- **Adaptive Sampling Algorithms:** Sample the uncertainty set $\mathcal{U}_{\hat{\theta}}$ efficiently at each iteration while solving the problem.
- **Closed-Form Expression:** Compute the supremum in closed form when possible.
- **Via Lagrange Duality:** Rewrite the worst-case supremum as an infimum that can be combined with the outer portfolio optimization layer.
- **Saddle-Point Optimization:** Use numerical methods designed for minimax problems or related saddle-point problems (Bertsekas 1999; Tütüncü and Koenig 2004).

Example: Worst-Case Constraint in Closed Form

- Quadratic constraint $f(\mathbf{x}; \boldsymbol{\theta}) = (\mathbf{c}^\top \mathbf{x})^2$, with parameter $\boldsymbol{\theta} = \mathbf{c}$.
- Belongs to a spherical uncertainty set:

$$\mathcal{U} = \{\mathbf{c} \mid \|\mathbf{c} - \hat{\mathbf{c}}\|_2 \leq \epsilon\}.$$

- Worst-case value:

$$\begin{aligned} \sup_{\mathbf{c} \in \mathcal{U}} |\mathbf{c}^\top \mathbf{x}| &= \sup_{\|\boldsymbol{\delta}\| \leq \epsilon} |(\hat{\mathbf{c}} + \boldsymbol{\delta})^\top \mathbf{x}| \\ &= |\hat{\mathbf{c}}^\top \mathbf{x}| + \sup_{\|\boldsymbol{\delta}\| \leq \epsilon} |\boldsymbol{\delta}^\top \mathbf{x}| \\ &= |\hat{\mathbf{c}}^\top \mathbf{x}| + \epsilon \|\mathbf{x}\|_2, \end{aligned}$$

where we have used the triangle inequality and Cauchy-Schwarz's inequality (upper bound achieved by $\boldsymbol{\delta} = \epsilon \mathbf{x} / \|\mathbf{x}\|_2$).

- Interpretation: the worst-case expression has the form of the naive version plus a regularization term.

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Worst-Case Mean Vector μ

Global Maximum Return Portfolio (GMRP): For an estimated mean vector $\hat{\mu}$:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \mathbf{w}^\top \hat{\mu} \\ & \text{subject to} && \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \end{aligned}$$

- Known to be highly sensitive to estimation errors.

Worst-Case Formulation: Instead of the naive GMRP, use the worst-case formulation:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \inf_{\mu \in \mathcal{U}_\mu} \mathbf{w}^\top \mu \\ & \text{subject to} && \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \end{aligned}$$

- Typical choices for the uncertainty region $\mathcal{U}_\mu$:
 - **Ellipsoidal set:**

$$\mathcal{U}_\mu = \left\{ \mu \mid (\mu - \hat{\mu})^\top \mathbf{S}^{-1} (\mu - \hat{\mu}) \leq \epsilon^2 \right\}.$$

- **Box set:**

$$\mathcal{U}_\mu = \{ \mu \mid \|\mu - \hat{\mu}\|_\infty \leq \epsilon \}.$$

Worst-Case Mean Vector μ

Worst-Case Mean Vector under Ellipsoidal Uncertainty Set

- **Ellipsoidal uncertainty set for μ :**

$$\mathcal{U}_\mu = \left\{ \mu = \hat{\mu} + \kappa \mathbf{S}^{1/2} \mathbf{u} \mid \|\mathbf{u}\|_2 \leq 1 \right\},$$

where:

- $\mathbf{S}^{1/2}$ is the symmetric square-root matrix of the shape $\mathbf{S}$, e.g., $\mathbf{S} = (1/T)\Sigma$;
 - κ determines the size of the ellipsoid.
- **Worst-case value of $\mathbf{w}^\top \mu$:**

$$\begin{aligned} \inf_{\mu \in \mathcal{U}_\mu} \mathbf{w}^\top \mu &= \inf_{\|\mathbf{u}\| \leq 1} \mathbf{w}^\top (\hat{\mu} + \kappa \mathbf{S}^{1/2} \mathbf{u}) \\ &= \mathbf{w}^\top \hat{\mu} + \kappa \inf_{\|\mathbf{u}\| \leq 1} \mathbf{w}^\top \mathbf{S}^{1/2} \mathbf{u} \\ &= \mathbf{w}^\top \hat{\mu} - \kappa \|\mathbf{S}^{1/2} \mathbf{w}\|_2. \end{aligned}$$

Worst-Case Mean Vector under Box Uncertainty Set

- **Box uncertainty set for μ :**

$$\mathcal{U}_\mu = \{\mu \mid -\delta \leq \mu - \hat{\mu} \leq \delta\},$$

where δ is the half-width of the box in all dimensions.

- **Worst-case value of $\mathbf{w}^\top \mu$:**

$$\inf_{\mu \in \mathcal{U}_\mu} \mathbf{w}^\top \mu = \mathbf{w}^\top \hat{\mu} - |\mathbf{w}|^\top \delta.$$

Worst-Case Mean Vector μ

Robust GMRP with Ellipsoidal Uncertainty Set:

- Robust version of the GMRP under ellipsoidal uncertainty set:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \mathbf{w}^T \hat{\mu} - \kappa \|\mathbf{S}^{1/2} \mathbf{w}\|_2 \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \end{aligned}$$

- Still a convex problem.
- Complexity increased to a second-order cone program (from a simple linear program in the naive formulation).

Robust GMRP with Box Uncertainty Set:

- Robust version of the GMRP under box uncertainty set:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{maximize}} && \mathbf{w}^T \hat{\mu} - |\mathbf{w}|^T \delta \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \end{aligned}$$

- Still a convex problem: can be rewritten as a linear program.
- Under constraints $\mathbf{1}^T \mathbf{w} = 1$ and $\mathbf{w} \geq \mathbf{0}$, the problem reduces to a naive GMRP where $\hat{\mu}$ is replaced by $\hat{\mu} - \delta$.

Quintile Portfolio as a Robust Portfolio

Other Uncertainty Sets:

- Various uncertainty sets can be considered, such as the ℓ_1 -norm ball.

Example: Quintile Portfolio as a Robust Portfolio

- **Quintile portfolio:**
 - A heuristic portfolio widely used by practitioners.
 - Selects the top fifth of the assets (or a different fraction) and equally allocates capital among them.
 - Common-sense heuristic portfolio.
- **Formal derivation as a robust portfolio:**
 - The quintile portfolio can be shown to be the optimal solution to the worst-case GMRP with an ℓ_1 -norm ball uncertainty set around the estimated mean vector (Zhou and Palomar 2020):

$$\mathcal{U}_\mu = \{\hat{\boldsymbol{\mu}} + \mathbf{e} \mid \|\mathbf{e}\|_1 \leq \epsilon\}.$$

Worst-Case Covariance Matrix Σ

Global Minimum Variance Portfolio (GMVP): For an estimated covariance matrix $\hat{\Sigma}$:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} && \mathbf{w}^T \hat{\Sigma} \mathbf{w} \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \end{aligned}$$

- Sensitive to estimation errors (though less so than errors in μ).

Worst-Case Formulation: Instead of the naive GMVP, use the worst-case formulation:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} && \sup_{\Sigma \in \mathcal{U}_{\Sigma}} \mathbf{w}^T \Sigma \mathbf{w} \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \end{aligned}$$

- Typical choices for the uncertainty region $\mathcal{U}_{\Sigma}$:

- **Spherical set:**

$$\mathcal{U}_{\Sigma} = \{ \Sigma \mid \|\Sigma - \hat{\Sigma}\|_F \leq \epsilon \}$$

- **Ellipsoidal set:**

$$\mathcal{U}_{\Sigma} = \{ \Sigma \mid \text{vec}(\Sigma - \hat{\Sigma})^T \mathbf{S}^{-1} \text{vec}(\Sigma - \hat{\Sigma}) \leq \epsilon^2 \}$$

- **Box set:**

$$\mathcal{U}_{\Sigma} = \{ \Sigma \mid \|\Sigma - \hat{\Sigma}\|_{\infty} \leq \epsilon \}$$

Worst-Case Covariance Matrix Σ

Worst-Case Covariance Matrix under a Data Spherical Uncertainty Set

- **Spherical uncertainty set for data matrix $\mathbf{X}$:**

$$\mathcal{U}_{\mathbf{X}} = \left\{ \mathbf{X} \mid \|\mathbf{X} - \hat{\mathbf{X}}\|_{\text{F}} \leq \epsilon \right\},$$

where ϵ determines the size of the sphere.

- **Worst-case value of $\sqrt{\mathbf{w}^{\text{T}} \Sigma \mathbf{w}}$:** (assuming $\hat{\Sigma} = \frac{1}{T} \hat{\mathbf{X}}^{\text{T}} \hat{\mathbf{X}}$)

$$\begin{aligned} \sup_{\mathbf{X} \in \mathcal{U}_{\mathbf{X}}} \sqrt{\mathbf{w}^{\text{T}} \left(\frac{1}{T} \mathbf{X}^{\text{T}} \mathbf{X} \right) \mathbf{w}} &= \sup_{\|\Delta\|_{\text{F}} \leq \epsilon} \frac{1}{\sqrt{T}} \left\| (\hat{\mathbf{X}} + \Delta) \mathbf{w} \right\|_2 \\ &= \frac{1}{\sqrt{T}} \left\| \hat{\mathbf{X}} \mathbf{w} \right\|_2 + \sup_{\|\Delta\|_{\text{F}} \leq \epsilon} \frac{1}{\sqrt{T}} \left\| \Delta \mathbf{w} \right\|_2 \\ &= \frac{1}{\sqrt{T}} \left\| \hat{\mathbf{X}} \mathbf{w} \right\|_2 + \frac{1}{\sqrt{T}} \epsilon \left\| \mathbf{w} \right\|_2. \end{aligned}$$

Worst-Case Covariance Matrix Σ

Worst-Case Covariance Matrix under an Ellipsoidal Uncertainty Set

- **Ellipsoidal uncertainty set for Σ :**

$$\mathcal{U}_{\Sigma} = \left\{ \Sigma \succeq \mathbf{0} \mid (\text{vec}(\Sigma) - \text{vec}(\hat{\Sigma}))^T \mathbf{S}^{-1} (\text{vec}(\Sigma) - \text{vec}(\hat{\Sigma})) \leq \epsilon^2 \right\},$$

where:

- $\text{vec}(\cdot)$ denotes the vec operator that stacks the matrix argument into a vector.
- matrix $\mathbf{S}$ gives the shape of the ellipsoid.
- ϵ determines the size.
- **Worst-case value of $\mathbf{w}^T \Sigma \mathbf{w}$:** Given by the (convex) SDP (Palomar 2025, chap. 14):

$$\begin{aligned} & \underset{\mathbf{Z}}{\text{minimize}} && \text{Tr} \left(\hat{\Sigma} (\mathbf{w} \mathbf{w}^T + \mathbf{Z}) \right) + \epsilon \left\| \mathbf{S}^{1/2} \left(\text{vec}(\mathbf{w} \mathbf{w}^T) + \text{vec}(\mathbf{Z}) \right) \right\|_2 \\ & \text{subject to} && \mathbf{Z} \succeq \mathbf{0}. \end{aligned}$$

Worst-Case Covariance Matrix Σ

Worst-Case Covariance Matrix under a Box Uncertainty Set

- **Box uncertainty set for Σ :**

$$\mathcal{U}_{\Sigma} = \left\{ \Sigma \succeq \mathbf{0} \mid \underline{\Sigma} \leq \Sigma \leq \overline{\Sigma} \right\},$$

where $\underline{\Sigma}$ and $\overline{\Sigma}$ denote the elementwise lower and upper bounds, respectively.

- **Worst-case value of $\mathbf{w}^T \Sigma \mathbf{w}$:** Given by the (convex) SDP (Lobo 2000):

$$\begin{aligned} & \underset{\overline{\Lambda}, \underline{\Lambda}}{\text{minimize}} && \text{Tr}(\overline{\Lambda} \overline{\Sigma}) - \text{Tr}(\underline{\Lambda} \underline{\Sigma}) \\ & \text{subject to} && \begin{bmatrix} \overline{\Lambda} - \underline{\Lambda} & \mathbf{w} \\ \mathbf{w}^T & 1 \end{bmatrix} \succeq \mathbf{0} \\ & && \overline{\Lambda} \succeq \mathbf{0}, \quad \underline{\Lambda} \succeq \mathbf{0}. \end{aligned}$$

Worst-Case Covariance Matrix Σ

Robust GMVP with Spherical Uncertainty Set:

- Robust version of the GMVP under a spherical uncertainty set for the data matrix:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} && \left(\left\| \hat{\mathbf{X}} \mathbf{w} \right\|_2 + \epsilon \left\| \mathbf{w} \right\|_2 \right)^2 \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}. \end{aligned}$$

- Still a convex problem.
- Complexity increased to a second-order cone program (from a simple QP).

Tikhonov Regularization:

- Heuristic similar to the robust GMVP:

$$\begin{aligned} & \underset{\mathbf{w}}{\text{minimize}} && \left\| \hat{\mathbf{X}} \mathbf{w} \right\|_2^2 + \epsilon \left\| \mathbf{w} \right\|_2^2 \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}. \end{aligned}$$

- Objective function can be rewritten as:

$$\mathbf{w}^T \frac{1}{T} \left(\hat{\mathbf{X}}^T \hat{\mathbf{X}} + \epsilon I \right) \mathbf{w}.$$

- Leads to a regularized sample covariance matrix.

Worst-Case Covariance Matrix Σ

Robust GMVP with Ellipsoidal Uncertainty Set:

- Robust version of the GMVP under an ellipsoidal uncertainty set for the covariance matrix:

$$\begin{aligned} & \underset{\mathbf{w}, \mathbf{W}, \mathbf{Z}}{\text{minimize}} && \text{Tr}(\hat{\Sigma}(\mathbf{W} + \mathbf{Z})) + \epsilon \left\| \mathbf{S}^{1/2} (\text{vec}(\mathbf{W}) + \text{vec}(\mathbf{Z})) \right\|_2 \\ & \text{subject to} && \begin{bmatrix} \mathbf{W} & \mathbf{w} \\ \mathbf{w}^T & 1 \end{bmatrix} \succeq \mathbf{0} \\ & && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0} \\ & && \mathbf{Z} \succeq \mathbf{0}. \end{aligned}$$

- Still a convex problem.
- Complexity increased to a semidefinite program (from a simple QP in the naive formulation).
- Note: the first matrix inequality is equivalent to $\mathbf{W} \succeq \mathbf{w}\mathbf{w}^T$ and, at an optimal point, it can be shown to be satisfied with equality $\mathbf{W} = \mathbf{w}\mathbf{w}^T$.

Worst-Case Mean Vector μ and Covariance Matrix Σ

Combined Worst-Case Mean-Variance Portfolio:

- Combines uncertainty in the mean vector μ and the covariance matrix Σ under the mean-variance portfolio formulation.
- For illustration, consider the mean-variance worst-case portfolio formulation under box uncertainty sets for μ and Σ .

Formulation:

$$\begin{aligned} & \underset{\mathbf{w}, \bar{\Lambda}, \underline{\Lambda}}{\text{maximize}} && \mathbf{w}^T \hat{\mu} - \|\mathbf{w}\|^T \delta - \frac{\lambda}{2} \left(\text{Tr}(\bar{\Lambda} \bar{\Sigma}) - \text{Tr}(\underline{\Lambda} \underline{\Sigma}) \right) \\ & \text{subject to} && \mathbf{1}^T \mathbf{w} = 1, \quad \mathbf{w} \geq \mathbf{0}, \\ & && \begin{bmatrix} \bar{\Lambda} - \underline{\Lambda} & \mathbf{w} \\ \mathbf{w}^T & 1 \end{bmatrix} \succeq \mathbf{0} \\ & && \bar{\Lambda} \geq \mathbf{0}, \quad \underline{\Lambda} \geq \mathbf{0}. \end{aligned}$$

- This is a convex semidefinite problem (SDP).

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Effectiveness of Robust Portfolio Formulations:

- Depends on the shape and size of the uncertainty region.
- Parameters must be properly chosen and tuned to the type of data and nature of the financial market.
- Risk of overfitting the model to the training data; extra care is needed.

Goal of Robust Design:

- Aim to make the solution more stable and less sensitive to error realization.
- Not necessarily to improve performance.
- Focus on gaining robustness rather than achieving the best performance in a given backtest compared to a naive design.

Evaluation of Robust Portfolios Under Errors in Mean Vector μ :

- We next consider robustness against errors in mean vector μ , which is paramount.
- Robustness for the covariance matrix Σ is less critical, but should be also assessed.

Sensitivity Analysis:

- Sensitivity of a robust portfolio under an ellipsoidal uncertainty set for the mean vector μ over six different realizations of the estimation error.
- Compared to naive portfolio design, it is more stable and less sensitive.
- Similar results can be observed with other variations in the robust formulation.

Performance Assessment:

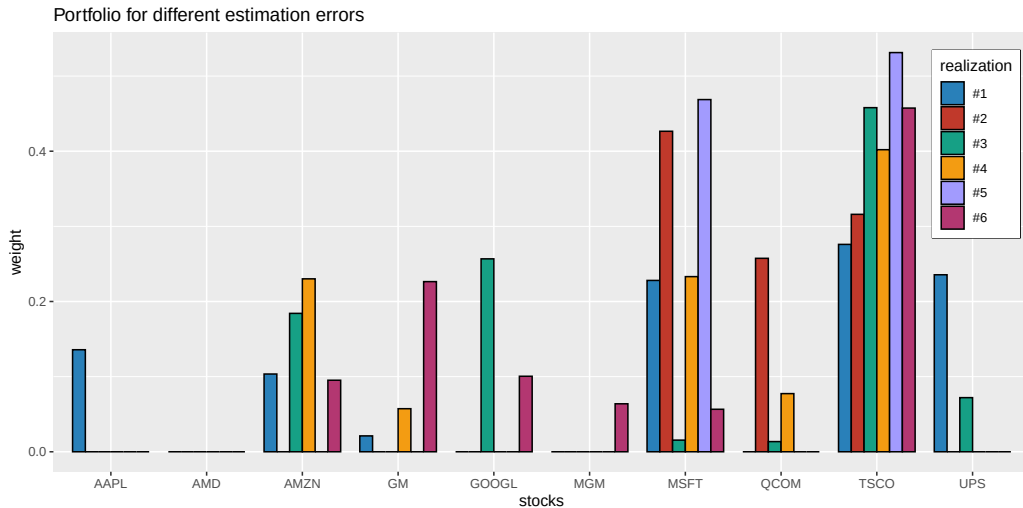
- Assess performance of mean-variance robust designs (with box and ellipsoidal uncertainty sets for the mean vector μ) compared to the naive design.
- Backtests conducted for 50 randomly chosen stocks from the S&P 500 during 2017-2020.

Empirical Distribution Analysis:

- Empirical distribution of the achieved mean-variance objective and the Sharpe ratio, calculated over 1,000 Monte Carlo noisy observations.
- Robust designs avoid the worst-case realizations (left tail in the distributions) at the expense of not achieving the best-case realizations (right tail).
- Robust designs are more stable and robust as expected.

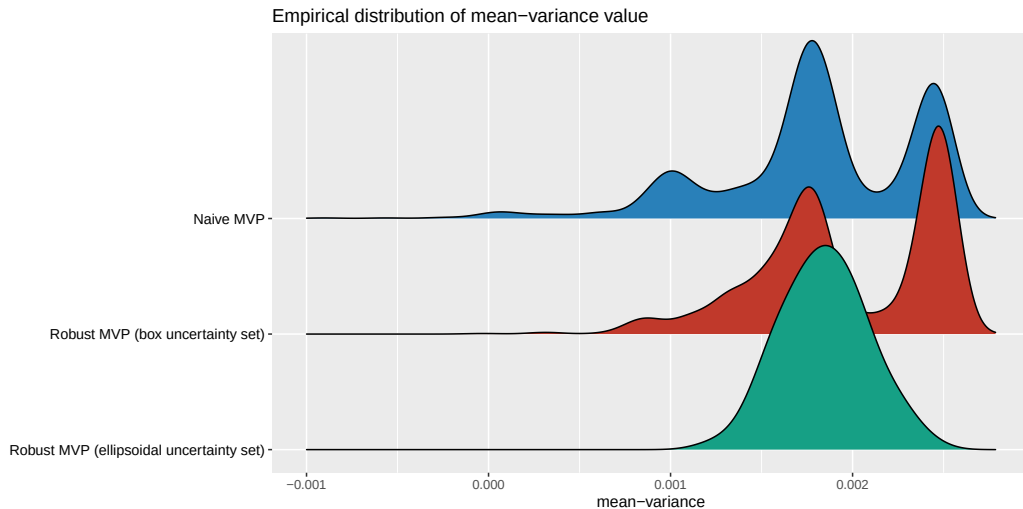
Numerical Experiments

Sensitivity of the robust mean–variance portfolio under an ellipsoidal uncertainty set for μ :



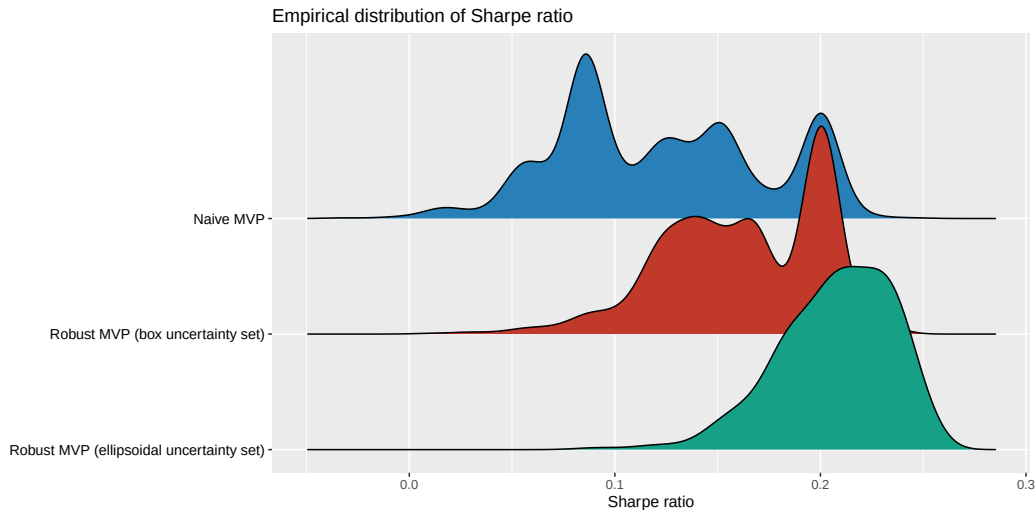
Numerical Experiments

Empirical performance distribution of naive versus robust mean–variance portfolios:



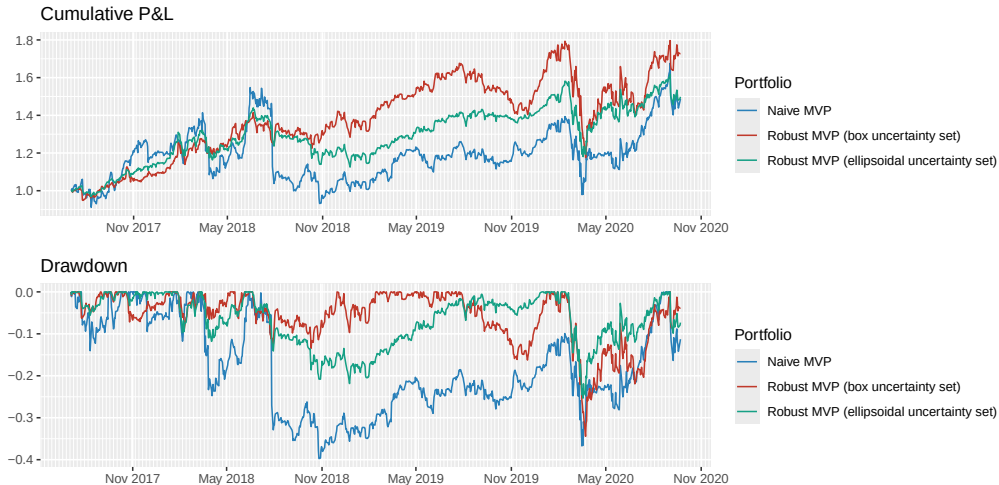
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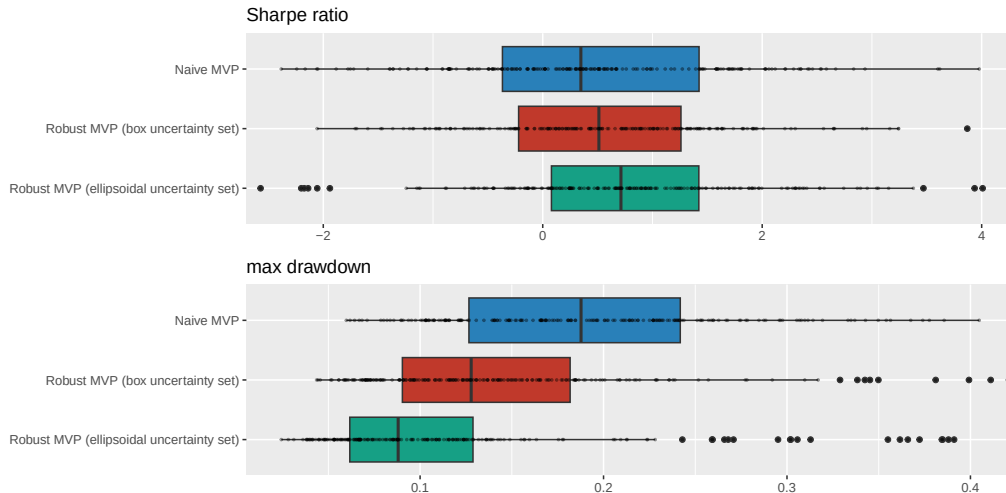
Numerical Experiments

Backtest of naive versus robust mean–variance portfolios:



Numerical Experiments

Multiple backtests of naive versus resampled mean–variance portfolios:



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- 1 Introduction
- 2 Robust Portfolio Optimization
 - Robust Optimization
 - Robust Worst-Case Portfolios
 - Numerical Experiments
- 3 Portfolio Resampling
 - Resampling Methods
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What Are Resampling Methods?

Importance of Estimating Parameter Accuracy:

- Estimating a parameter θ is of little use without knowing the accuracy of the estimate.
- Confidence intervals are key in statistical inference, allowing localization of the true parameter with a certain confidence level (e.g., 95%).
- Traditionally, confidence intervals were derived using theoretical mathematics.
- Resampling methods use computer-based numerical techniques to assess statistical accuracy without formulas (Efron and Tibshirani 1993).

Resampling Methods:

- Creation of new samples based on a single observed sample block.
- Suppose we have n observations, $\mathbf{x}_1, \dots, \mathbf{x}_n$, of a random variable $\mathbf{x}$ from which we estimate some parameters θ as:

$$\hat{\theta} = f(\mathbf{x}_1, \dots, \mathbf{x}_n)$$

- The estimation $\hat{\theta}$ is a random variable because it is based on n random variables.
- Resampling methods help characterize the distribution of the estimation without needing more realizations of the random variable $\mathbf{x}$.

Cross-Validation:

- Widely used in portfolio backtesting and machine learning.
- Divides n observations into two groups: a training set for fitting the estimator $f(\cdot)$ and a validation set for assessing its performance.
- Repeated multiple times to provide multiple realizations of the performance value.
- **k -fold cross-validation:** divides the set into k subsets, each held out in turn as the validation set while using the others for training.
- **Leave-one-out cross-validation:** extreme case where the original dataset of n observations is divided into $k = n$ subsets, with each subset holding out a single observation for validation.

The Bootstrap:

- Proposed in 1979 by Efron, based on sound statistical theory (Efron 1979).
- Mimics the original sampling process by sampling n times with replacement from the original n observations.
- Repeated B times to obtain bootstrap samples:

$$(\mathbf{x}_1, \dots, \mathbf{x}_n) \rightarrow (\mathbf{x}_1^{*(b)}, \dots, \mathbf{x}_n^{*(b)}), \quad b = 1, \dots, B.$$

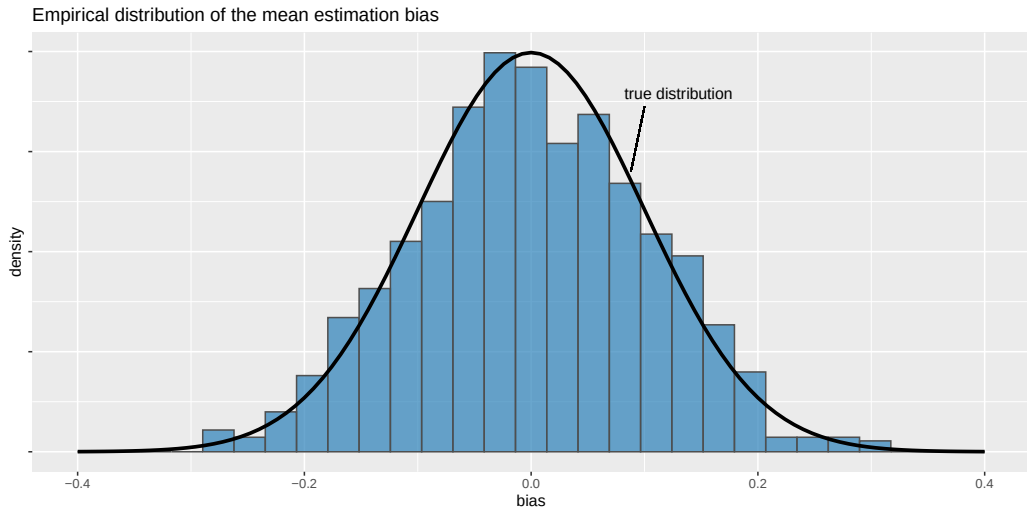
- Leads to different realizations of the estimation (bootstrap replicates):

$$\hat{\theta}^{*(b)} = f(\mathbf{x}_1^{*(b)}, \dots, \mathbf{x}_n^{*(b)}), \quad b = 1, \dots, B.$$

- Measures of accuracy (bias, variance, confidence intervals, etc.) can be empirically obtained.
- Theoretical result: statistical behavior of the random resampled estimates $\hat{\theta}^{*(b)}$ compared to $\hat{\theta}$ faithfully represents the statistics of the random estimates $\hat{\theta}$ compared to the true parameter θ (Efron and Tibshirani 1993).
- Estimations of accuracy are asymptotically consistent as $B \rightarrow \infty$.

Resampling Methods

Empirical distribution of the sample mean bias via the bootstrap:



Resampling Methods

Several variations and extensions of the basic bootstrap have been developed over the years:

Parametric Bootstrap:

- Assumes a specific distribution for the data (e.g., Gaussian) (Efron and Tibshirani 1993).
- Steps:
 - ① Assume a parametric distribution for the data.
 - ② Estimate the distribution parameters from the observed data.
 - ③ Generate data using the estimated parametric distribution.

Block Bootstrap:

- Addresses structural dependency in data (Lahiri 1999).
- Involves resampling blocks of data rather than individual observations.

Bag of Little Bootstraps:

- Designed for large datasets with many observations (Kleiner et al. 2014).
- Steps:
 - ① Divide the dataset into smaller subsets.
 - ② Apply the bootstrap method to each subset.
 - ③ Aggregate results to assess estimator quality.

The Jackknife:

- Proposed in the mid-1950s by M. Quenouille.
- Derived for estimating biases and standard errors of sample estimators.
- Given n observations $\mathbf{x}_1, \dots, \mathbf{x}_n$, the i th jackknife sample is obtained by removing the i th data point:

$$\mathbf{x}_1, \dots, \mathbf{x}_{i-1}, \mathbf{x}_{i+1}, \dots, \mathbf{x}_n.$$

- Produces $B = n$ bootstrap samples each with $n - 1$ observations.
- An approximation to the bootstrap; makes a linear approximation to the bootstrap.
- Accuracy depends on how “smooth” the estimator is; for highly nonlinear functions, the jackknife can be inefficient.

Bagging (Bootstrap Aggregating):

- Method for generating multiple versions of an estimator or predictor via the bootstrap and then using these to get an aggregated version (Breiman 1996; Hastie, Tibshirani, and Friedman 2009).
- Improves accuracy of the basic estimator or predictor, which typically suffers from sensitivity to the realization of the random data.
- Mathematically, bagging is a simple average of the bootstrap replicates:

$$\hat{\theta}^{\text{bag}} = \frac{1}{B} \sum_{b=1}^B \hat{\theta}^{*(b)}.$$

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Portfolio Resampling

Optimization Problem in Portfolio Design:

- Formulated based on T observations of assets' returns $\mathbf{x}_1, \dots, \mathbf{x}_T$.
- Solution is an optimal portfolio $\mathbf{w}$.
- Sensitive to noise in observed data and estimated parameters ($\hat{\boldsymbol{\mu}}$ and $\hat{\boldsymbol{\Sigma}}$).

Resampling Techniques:

- Utilize statistical methods from the past half-century.
- Techniques include bootstrap and bagging.
- Improve portfolio design by reducing sensitivity to data noise.

Historical Context:

- Resampling proposed in the 1990s to assess portfolio accuracy.
- Naive approach: use available data to design mean-variance portfolios and obtain the efficient frontier.
- Issue: the computed efficient frontier is unreliable due to high sensitivity.

Resampled Efficient Frontier: (Jorion 1992; Michaud and Michaud 1998)

- Resampling allows computation of a more reliable efficient frontier.
- Identifies statistically equivalent portfolios.

Portfolio Bagging

Aggregating Portfolios via Bagging: Technique considered in 1998 for portfolio aggregation using a bagging procedure (Michaud and Michaud 1998, 2008; Scherer 2002).

Steps in the Bagging Procedure:

- 1 **Resample the original data:** resample the original data $(\mathbf{x}_1, \dots, \mathbf{x}_T)$ B times using the bootstrap method and estimate B different versions of the mean vector and covariance matrix: $\hat{\boldsymbol{\mu}}^{*(b)}$ and $\hat{\boldsymbol{\Sigma}}^{*(b)}$.
- 2 **Solve the optimal portfolio:** solve the optimal portfolio $\mathbf{w}^{*(b)}$ for each bootstrap sample.
- 3 **Average the portfolios via bagging:**

$$\mathbf{w}^{\text{bag}} = \frac{1}{B} \sum_{b=1}^B \mathbf{w}^{*(b)}.$$

Observations:

- The bagging procedure for portfolio aggregation is straightforward.
- The main bottleneck is the increase in computational cost by a factor of the number of bootstraps B compared to the naive approach.

Portfolio Subset Resampling

Idea: Sample the asset dimension rather than the observation (temporal) dimension.

Procedure:

① **Select subset size:**

- Choose the subset size using the rule-of-thumb $\lceil N^{0.7} \rceil$ or $\lceil N^{0.8} \rceil$.
- Example: For $N = 50$, subset sizes would be 16 or 23 assets, respectively.

② **Generate random subset:**

- Randomly select a subset of assets according to the chosen size.
- This reduces the portfolio dimensionality and computational cost.

③ **Design portfolio:**

- Solve the portfolio optimization problem using only the selected subset of assets.
- The resulting portfolio has weights only for the selected assets (other weights are zero).

④ **Repeat:**

- Repeat steps 2-3 multiple times with different random subsets.

⑤ **Aggregate portfolios:**

- Average all computed portfolios to obtain the final portfolio.
- Since each portfolio is sparse (non-zero weights only for selected assets), the averaging naturally handles the different dimensions.

Benefits:

- **Reduced computational cost:** significant reduction in computational cost due to smaller subset size.
- **Improved parameter estimation:** better estimation of parameters because the ratio of observations-to-dimensionality has increased.
 - Example: for $T = 252$ daily observations and $N = 50$ assets, the nominal ratio would be $T/N \approx 5$, but the subset resampling ratio is $T/N^{0.7} \approx 16$ or $T/N^{0.8} \approx 11$.
 - Higher ratio leads to more reliable parameter estimates.

Numerical Experiments:

- Subsequent experiments will show that subset resampling is an effective technique in practice.

Combination of Techniques:

- Combine random subset resampling along the asset domain with the bootstrap along the temporal domain (Shen et al. 2019).
- Each bootstrap sample contains only a subset of the N assets.
- This combination leverages the benefits of both resampling techniques to create more robust and computationally efficient portfolios.

Summary:

- **Computational savings and improved robustness:** portfolio subset resampling and portfolio subset bagging achieve significant computational savings and improve the robustness and reliability of portfolios.
- **Mitigating sensitivity to noise:** help mitigate sensitivity to noise and variability in the data, leading to more stable investment strategies.

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Sensitivity Analysis:

- **Naive mean-variance portfolio (MVP):** extremely sensitive to estimation errors as previously seen.
- **Robust portfolio optimization:** less sensitive to estimation errors as shown next as previously seen.
- **Bagged portfolios:** more stable and less sensitive with $B = 200$ bootstrap samples.

Performance Assessment:

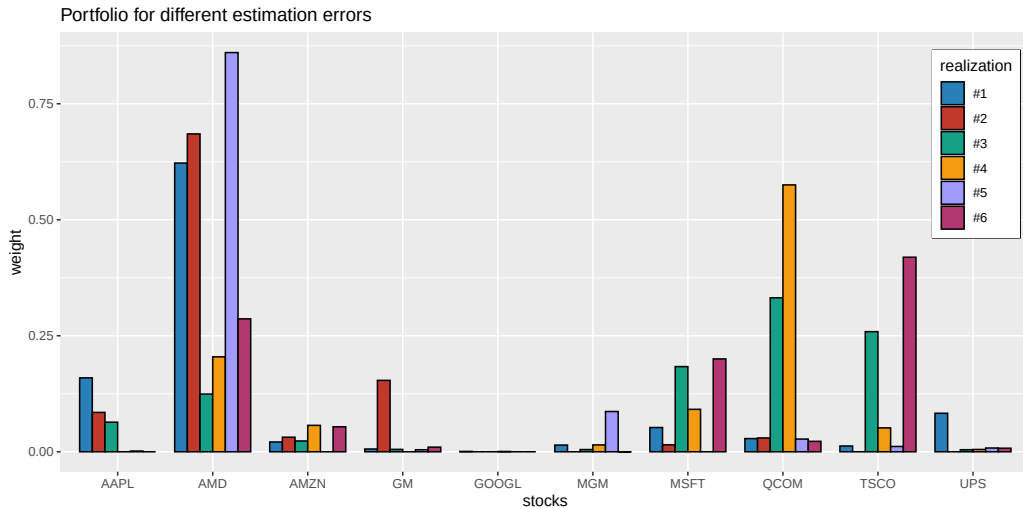
- Assess performance of resampled portfolios (bagging, subset resampling, subset bagging) compared with naive design.
- Backtests conducted for 50 random stocks from the S&P 500 during 2017-2020.

Empirical Distribution Analysis:

- Empirical distribution of the achieved mean-variance objective and the Sharpe ratio, calculated over 1,000 Monte Carlo noisy observations.
- Resampled portfolios are more stable, avoid extreme bad realizations (although the naive design can be superior in some cases).
- Resampled portfolios seem to be superior in Sharpe ratio and drawdown.

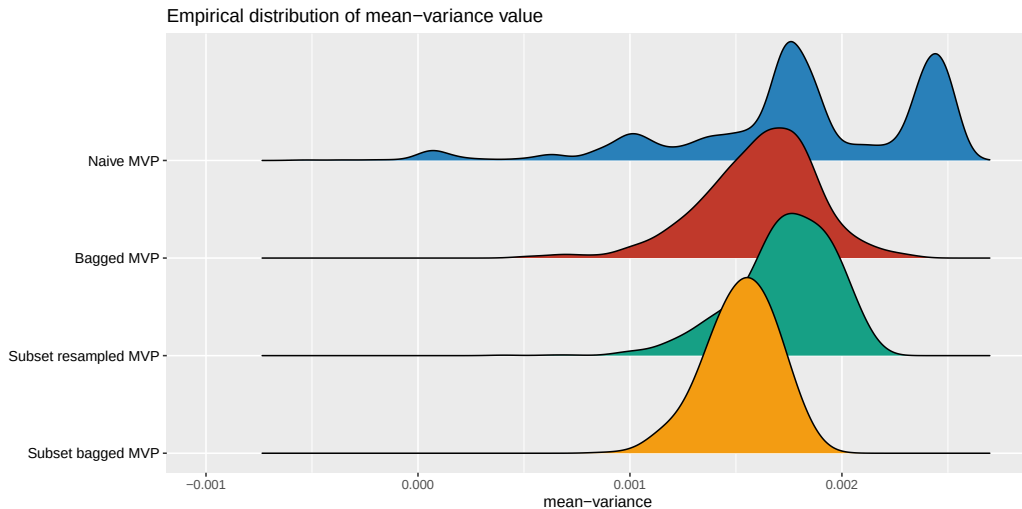
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Sensitivity of the bagged mean–variance portfolio:



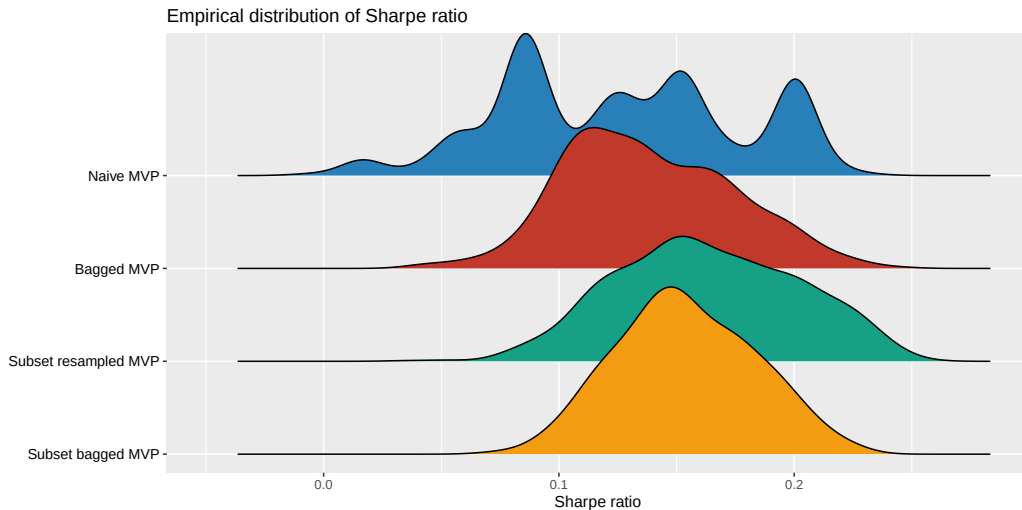
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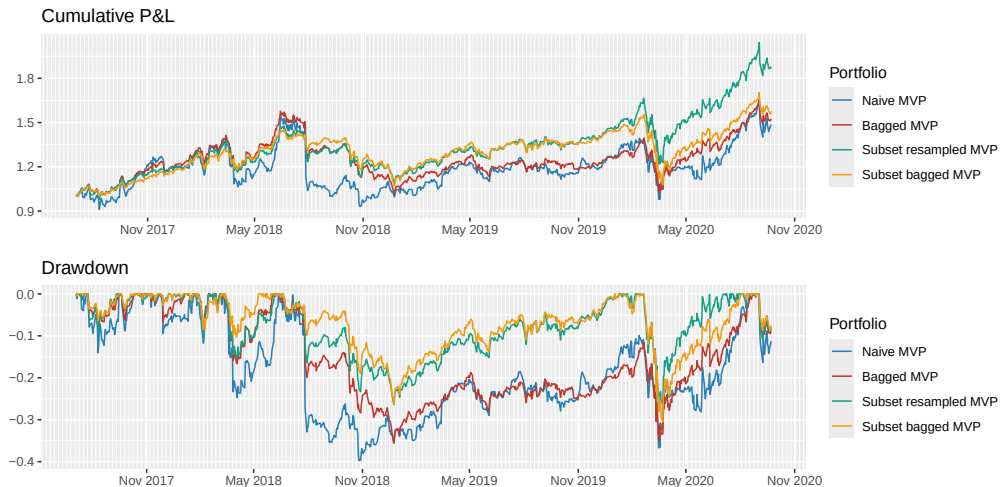
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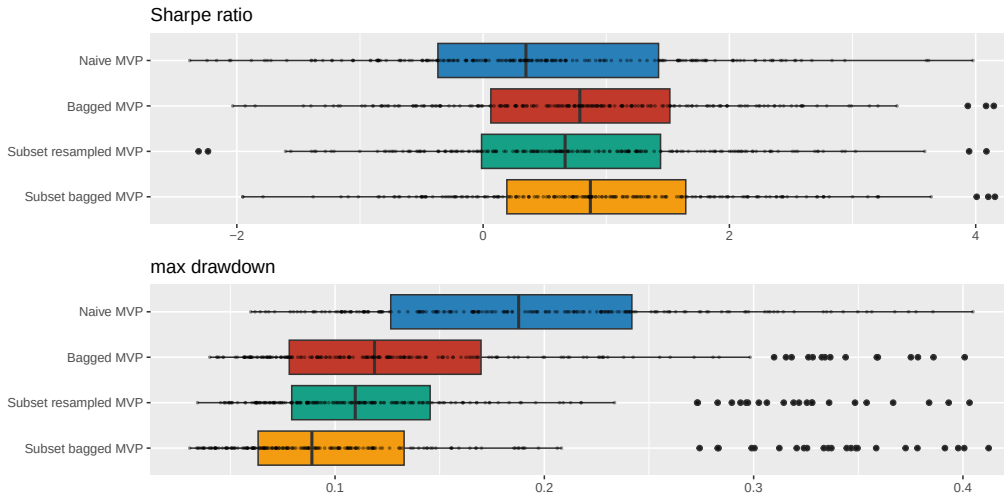
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Summary

- Ideally, a portfolio optimization solution should meet the desired objective within constraints.
- In practice, it often fails due to reliance on estimated parameters like the mean vector and covariance matrix, which contain errors from noisy and limited data.
- Ignoring these estimation errors can lead to disastrous results, earning the term “estimation-error maximizers” for such portfolio problems.
- Effective approaches to mitigate naive solutions include:
 - **Robust portfolios:** Incorporate parameter errors using robust optimization, a well-developed method in portfolio optimization.
 - **Resampled portfolios:** Use bootstrapping and resampling to aggregate multiple naive solutions into a more stable and reliable portfolio.

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